

ES -116

Principles and Applications of Electrical Engineering

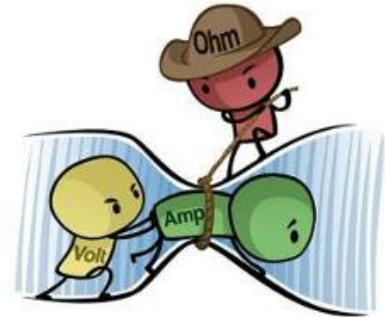
Semester II, 2025-26

Transitioning to Non-Linear Circuits With Diodes

Recap of Linear components: R, L and C ($V=IZ$)

1. Resistor (The Dissipator)

- The Reliable Friend. They don't have mood swings. Whether you talk fast (High Freq.) or slow (Low Freq.), they treat you exactly the same.
- Dissipates energy (turns it into heat). *Motto*: I burn what I take.
- Frequency: Flat Response. It resists current equally at all frequencies.



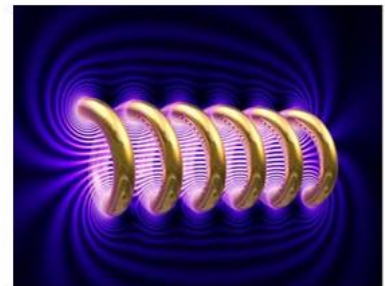
2. Capacitor (The Tank)

- The Secret Keeper. It holds onto your secrets (Charge) and supports you (Voltage) when you feel drained. It takes time to open up (Charging time).
- Stabilizes Voltage (opposes changes in V). Stores energy in the Electric Field.
- *Blocks Low Freq (DC)*: Acts like a wall. *Passes High Freq (AC)*: Acts like a short circuit.
- *Motto*: I block the constant, but let the changes pass.

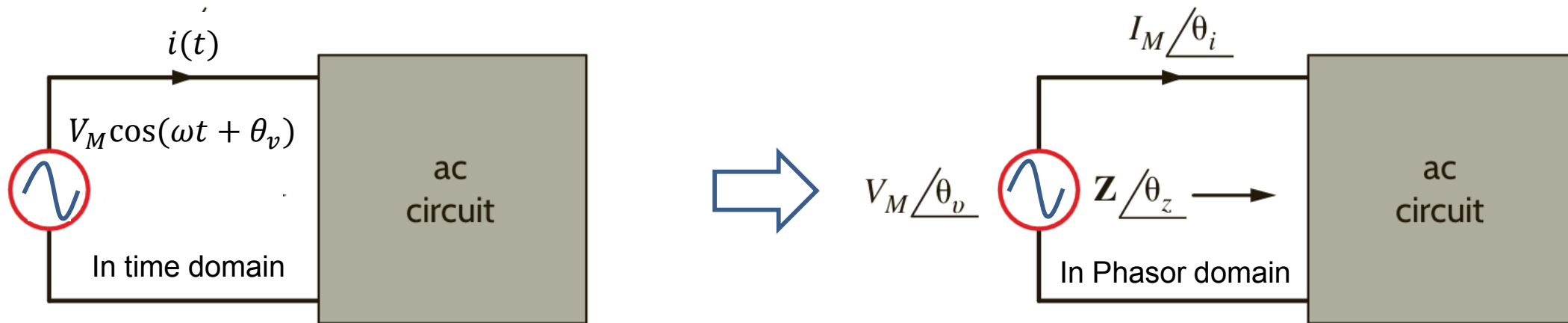


3. Inductor (The Flywheel)

- The Heavy Friend with an Aura. It stores energy in its magnetic aura (Field). It hates being rushed i.e. if you try to change current instantly, it fights back!
- Stabilizes Current (opposes changes in I). Stores energy in the Magnetic Field.
- *Passes Low Freq (DC)*: Acts like a wire, *Blocks High Freq (AC)*: Acts like a wall. *Motto*: I like the steady flow; don't push me too fast.



Recap: Linear circuits with R, L and C with AC input

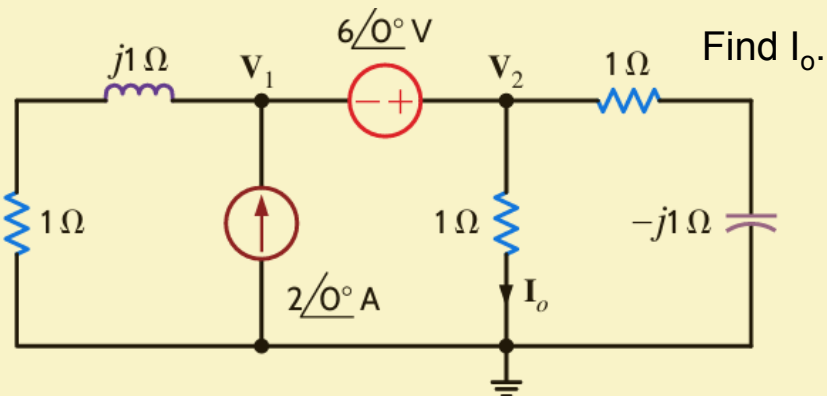


$$\mathbf{Z} = \frac{V_M \angle \theta_v}{I_M \angle \theta_i} = \frac{V_M}{I_M} \angle \theta_v - \theta_i = Z \angle \theta_z \quad \mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{\mathbf{I}}{\mathbf{V}}$$

$$Z \angle \theta_z = R + jX$$

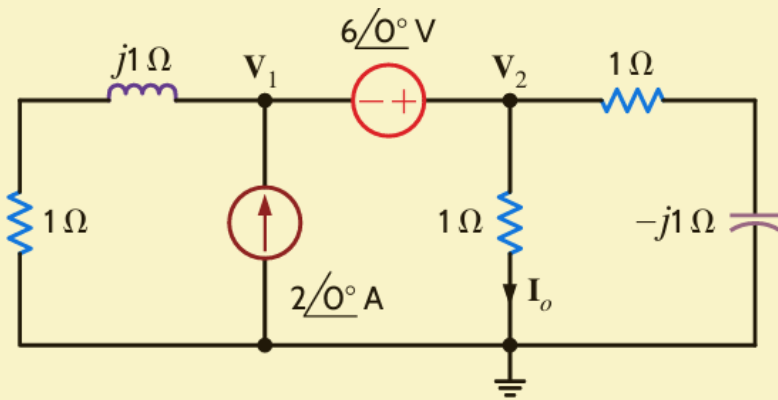
$$Z = \sqrt{R^2 + X^2}$$

$$\theta_z = \tan^{-1} \frac{X}{R}$$



Recap: Linear circuits with R, L and C

Problem 8.15 in Irwin book's chapter (AC STEADY-STATE ANALYSIS), find I_o .



Using Nodal Analysis:

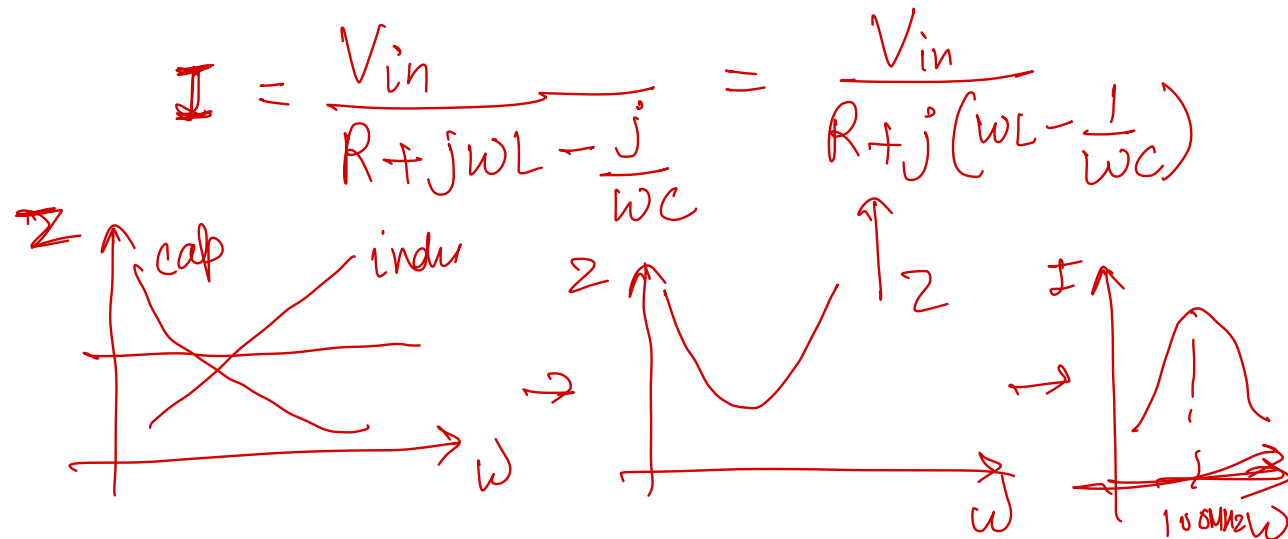
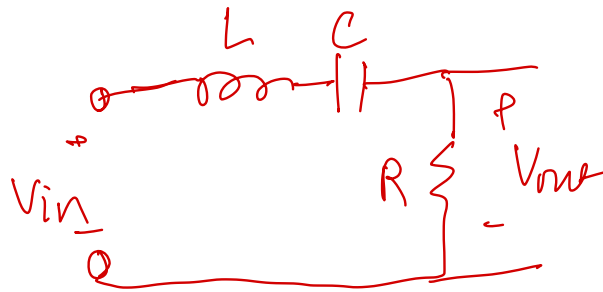
$$\begin{pmatrix} 0.5 - 0.5j & 1.5 + 0.5j \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}$$

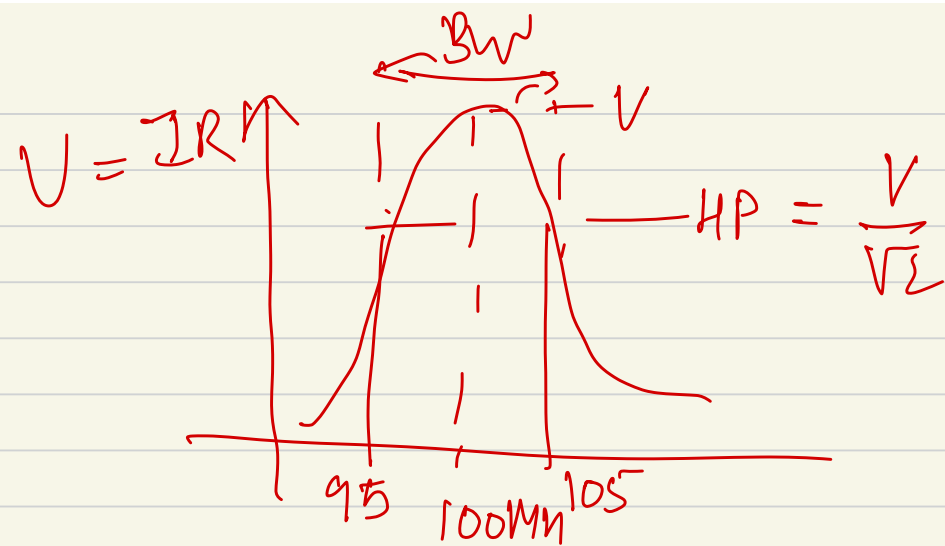
Linear set of equations converted to the Matrix form: $\mathbf{YV} = \mathbf{I}$

Candy Problem: Trying to Solve with R, L and C

You are trying to listen to Campus Radio at 100 MHz, but your antenna is picking up static from a Classic Rock station at 90 MHz and a Pop station at 110 MHz.

Q. Design a circuit using R, L and C that allows the 100 MHz signal to pass through to the speaker while blocking the other two frequencies. We can aim for a Bandwidth of 10 MHz (95 MHz to 105 MHz). You have a 10 pF Capacitor in your kit. You need to calculate the required Inductor (L) and Resistor (R).

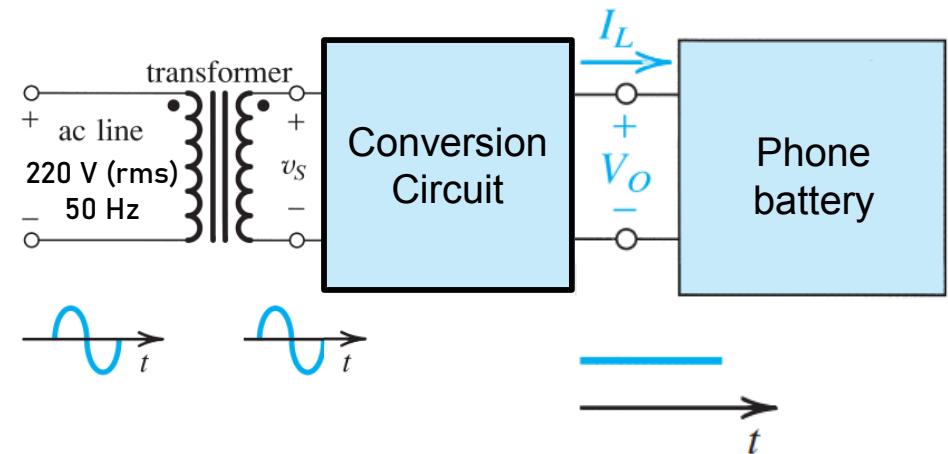




Another Problem: Trying to Solve with R, L and C

You would like to charge your phone using wall socket that provides you a nice AC signal. Your phone battery needs 5 V DC to charge. Therefore, we need a circuit in between to convert the AC signal from wall socket to a DC signal required for the phone battery.

Can you build this circuit using R and/or L and/or C?



Another Problem: Trying to Solve with R, L and C

You would like to charge your phone using wall socket that provides you a nice AC signal. Your phone battery needs 5 V DC to charge. Therefore, we need a circuit in between to convert the AC signal from wall socket to a DC signal required for the phone battery.

- What if we just use a voltage divider to drop the 220V AC down to 5V AC?
- *Result:* For the first half-cycle, current flows *into* the battery (Charging). But for the second half-cycle, the voltage flips! Current flows *out* of the battery (Discharging).
 - *Net Result:* Zero Charge. You are just heating up the battery.

- What if we use an Inductor to smooth it out?
- *Result:* Same problem. Inductors and Capacitors store energy, but they return it. They don't stop the "backflow."

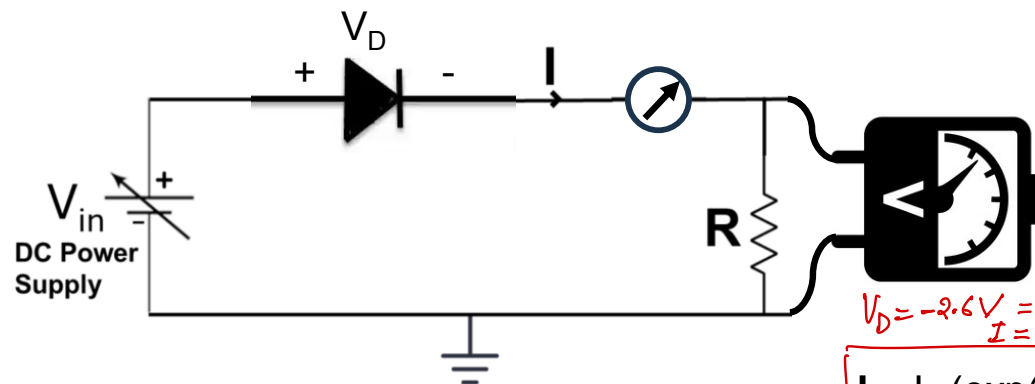
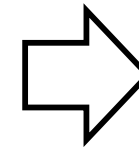
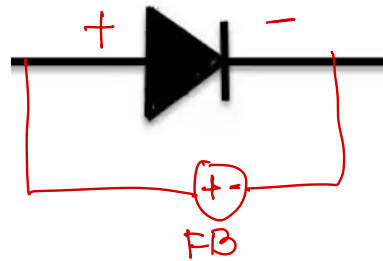
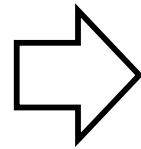
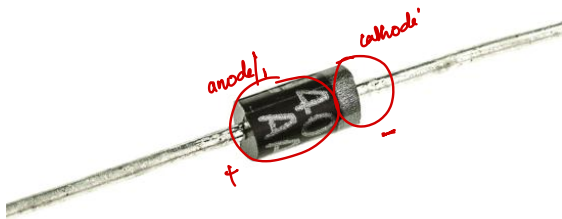
- We need to stop the "backflow". We need a component that says: 'Positive current? Come on in. Negative current? **STOP**.'
- This behavior blocking current in one direction while passing it in the other is impossible for R, L, and C because they are linear, bilateral components.
- To charge this phone, we need a **Non-Linear** device. We need a one-way street. We need a check valve.

An new non-linear component

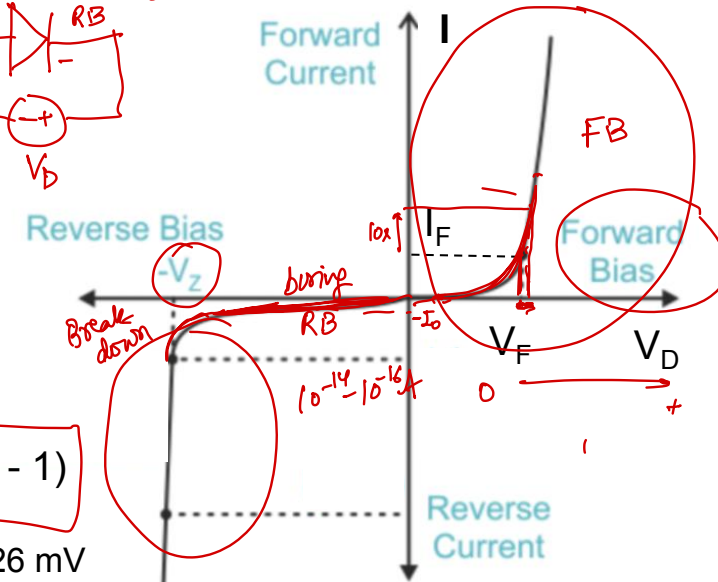
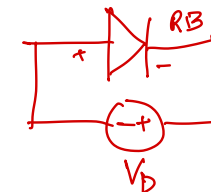
Diode: A two-terminal electronic device

Representation of the diode

Diode characteristics



non-linear, asymmetric



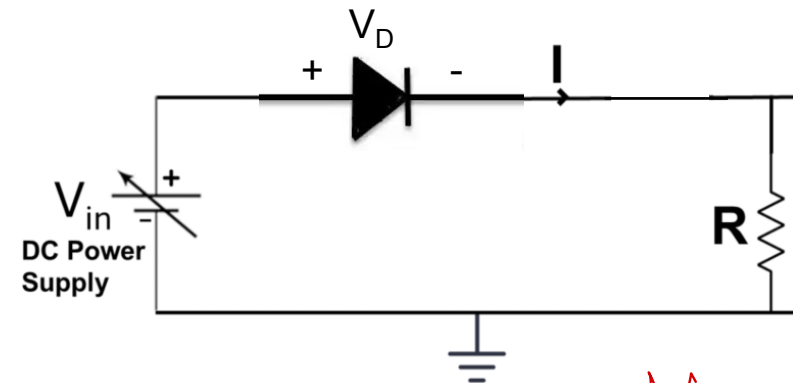
$$V_D = -2.6V = -100V_t$$

$$I = -I_0$$

$$I = I_0 (\exp(V_D/V_t) - 1)$$

thermal voltage
Here, $V_t = kT/q \sim 26 \text{ mV}$
@ Room Temp. (300K)

Let's solve it mathematically



$$I = I_0 (\exp(V_D/V_t) - 1)$$

Here, $V_t = 26 \text{ mV}$, $I_0 = 10^{-14} \text{ A}$.

For $V_{in} = 10 \text{ V DC}$, what will be the current flowing in the circuit if $R = 100 \Omega$?

$$V_D = V_{in} - V_R = V_{in} - IR$$

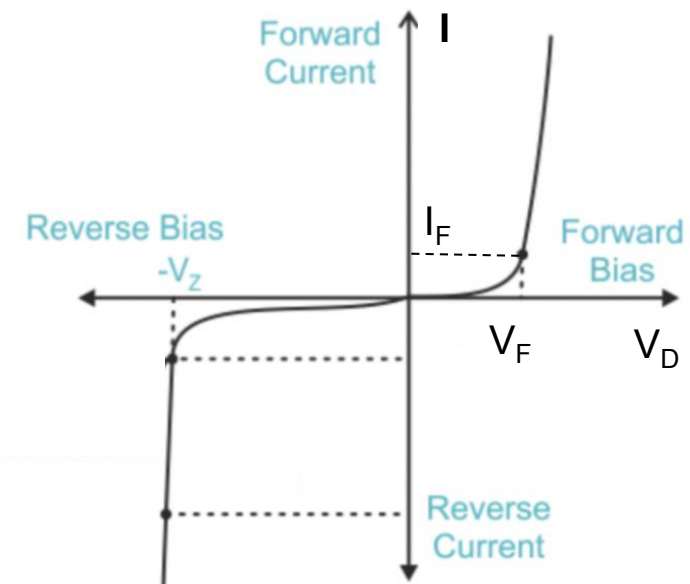
$$I = I_0 (e^{V_D/V_t} - 1)$$

$$V_D = V_{in} - I_0 R e^{V_D/V_t} + I_0 R$$

Implicit eq

can not be solved analytically

⇒ require 'hit & trial method' or iterative method.



Initial Guess 1:- $V_D = 0$ (Bad guess) $I_0 = 10^{-14} \text{ A}$

$$I = \frac{10}{100} = 100 \text{ mA}$$

$\ln \rightarrow \log_e$

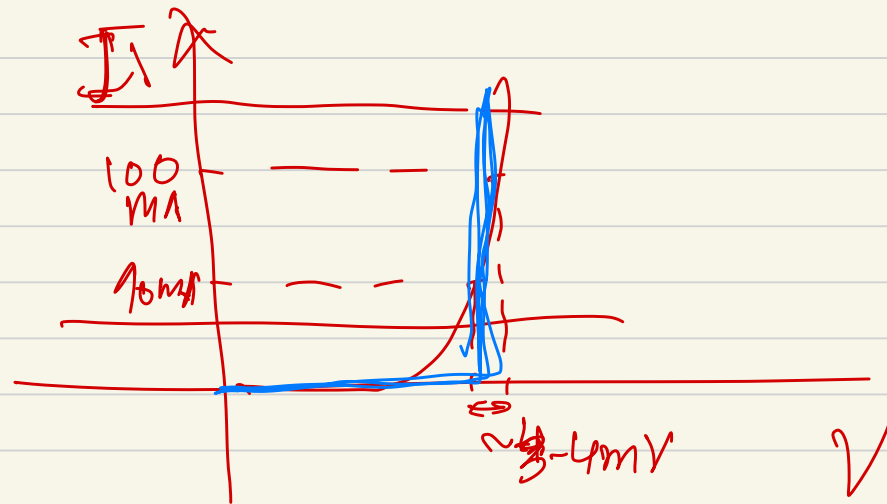
$$\begin{aligned} I &= I_0 (e^{V_D/V_T} - 1) \Rightarrow V_D = V_T \left(\ln \left(\frac{I}{I_0} + 1 \right) \right) \\ &= 25 \text{ mV} \cdot \left(\ln \left(\frac{100 \times 10^{-3}}{10^{-14}} + 1 \right) \right) \\ &= 25 \times 10^{-3} \ln(10^{13}) \\ &= \underline{0.748 \text{ V}} \end{aligned}$$

Guess 2: $V_D = 0.748 \text{ V}$

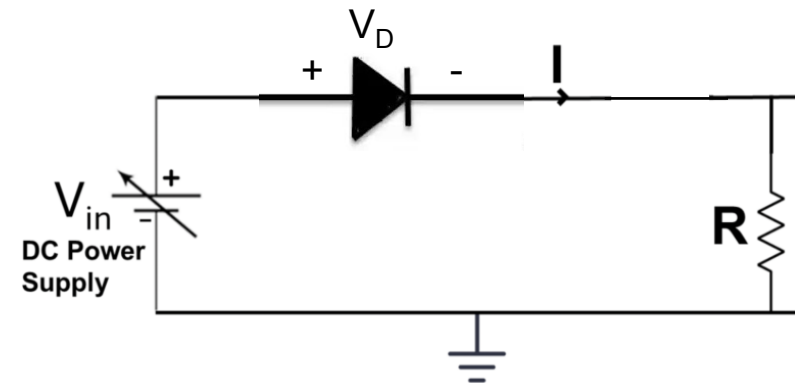
$$I = \frac{10 - 0.748}{100} = \frac{9.252}{100} = \underline{\underline{92.52 \text{ mA}}}$$

$$V_D = V_T \ln \left(\frac{92.52 \times 10^{-3}}{10^{-14}} \right) \approx 0.746 \text{ V}$$

$$V_D \approx 0.745 \text{ V} \rightarrow I =$$



Constant Voltage Drop Model



Using the constant voltage drop model, what will be the current flowing in the circuit for $R = 100 \Omega$ and $V_{in} = 10 \text{ V DC}$?

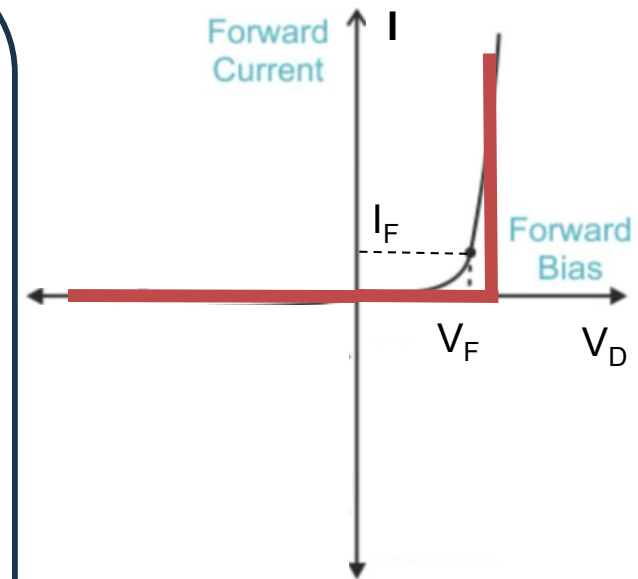
$$V_D = 0.7 \text{ V}, \quad I = \frac{10 - 0.7}{100} = 93 \text{ mA}$$

➤ How good is this approximation?

$$V_D = V_{\text{drop}}$$

How do we choose V_{drop} given the diode characteristics?
Does current range matter?

yes

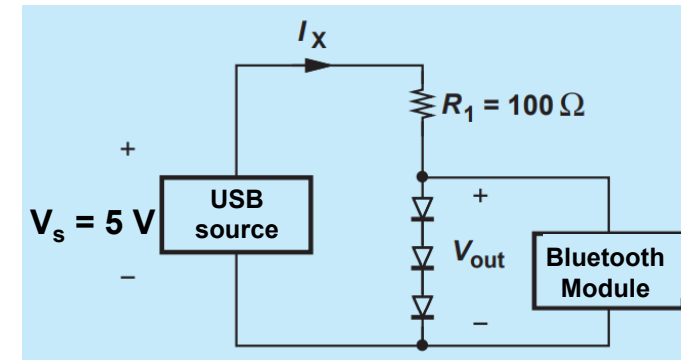


1. For widely used silicon p-n junction diodes,
 $V_{\text{drop}} = 0.55 - 0.7 \text{ V}$ for $I < 1 \text{ mA}$
and $V_{\text{drop}} = 0.8 - 1 \text{ V}$ for $I > 100 \text{ mA}$
2. Ge diodes: $V_{\text{drop}} = 0.2 - 0.3 \text{ V}$,
3. Light emitting diodes (LEDs): $V_{\text{drop}} = 1.5 - 2.5 \text{ V}$.
4. Schottky diodes: $V_{\text{drop}} = 0.15 - 0.2 \text{ V}$

Example (Optional)

You are building a portable gadget that uses a Bluetooth Module. This module requires a supply voltage of approximately **2.1 V** to operate. However, you only have a **5 V** USB power bank available. Instead of buying an expensive regulator chip, you decide to use your knowledge of diodes to construct a simple "Shunt Regulator" using standard silicon diodes ($V_{\text{drop}} \sim 0.7\text{V}$) and a current-limiting resistor R_1 .

- Q1. Calculate the diode current when the output is unloaded (Open Circuit).
Q2. Calculate the voltage across diodes, when bluetooth module is connected in parallel with diodes, drawing 15 mA.
Q3. Determine the percentage drop in voltage (Load Regulation) to demonstrate that the 'constant' 0.7 V drop is actually dependent on the current flowing through the diode.



Diode data sheet (IN 4001-4007)



Features

- Diffused Junction
- Low Forward Voltage Drop
- High Current Capability
- High Reliability
- High Surge Current Capability

Mechanical Data

- Case: DO-41, Molded Plastic
- Terminals: Plated Leads Solderable per MIL-STD-202, Method 208
- Polarity: Cathode Band
- Weight: 0.35 grams (approx.)
- Mounting Position: Any
- Marking: Type Number
- **Lead Free: For RoHS / Lead Free Version, Add "-LF" Suffix to Part Number, See Page 4**

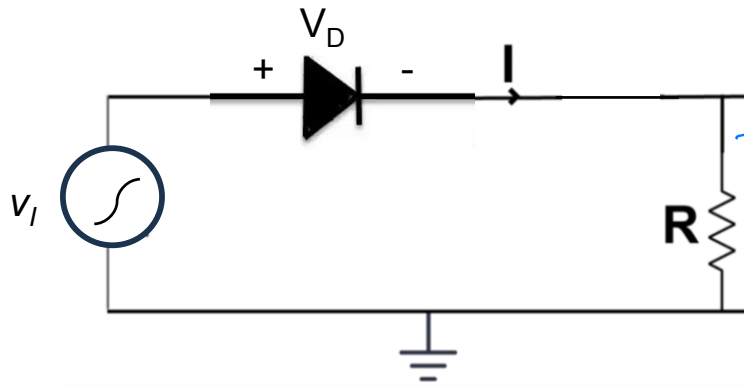
Maximum Ratings and Electrical Characteristics @T_A=25°C unless otherwise specified

Single Phase, half wave, 60Hz, resistive or inductive load.
For capacitive load, derate current by 20%.

Characteristic	Symbol	1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V _{RRM} V _{RWM} V _R	50	100	200	400	600	800	1000	V
RMS Reverse Voltage	V _{R(RMS)}	35	70	140	280	420	560	700	V
Average Rectified Output Current (Note 1) @T _A = 75°C	I _O	1.0							A
Non-Repetitive Peak Forward Surge Current 8.3ms Single half sine-wave superimposed on rated load (JEDEC Method)	I _{FSM}	30							A
Forward Voltage @I _F = 1.0A	V _{FM}	1.0							V
Peak Reverse Current @T _A = 25°C At Rated DC Blocking Voltage @T _A = 100°C	I _{RM}	5.0 50							μA
Typical Junction Capacitance (Note 2)	C _j	15							pF
Typical Thermal Resistance Junction to Ambient (Note 1)	R _{θJA}	50							°C/W
Operating Temperature Range	T _j	-65 to +125							°C
Storage Temperature Range	T _{STG}	-65 to +150							°C

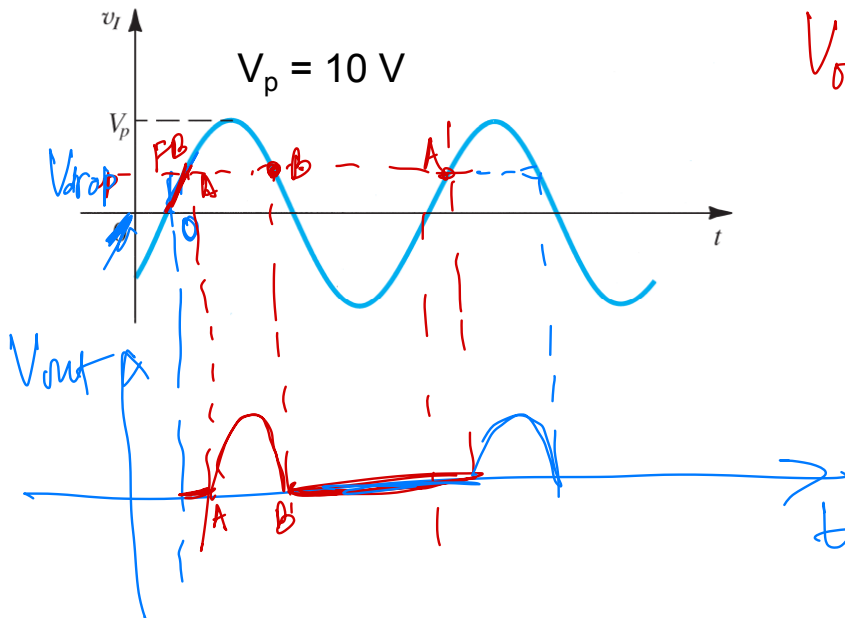
Note: 1. Leads maintained at ambient temperature at a distance of 9.5mm from the case
2. Measured at 1.0 MHz and Applied Reverse Voltage of 4.0V D.C.

Using constant Voltage Drop Model

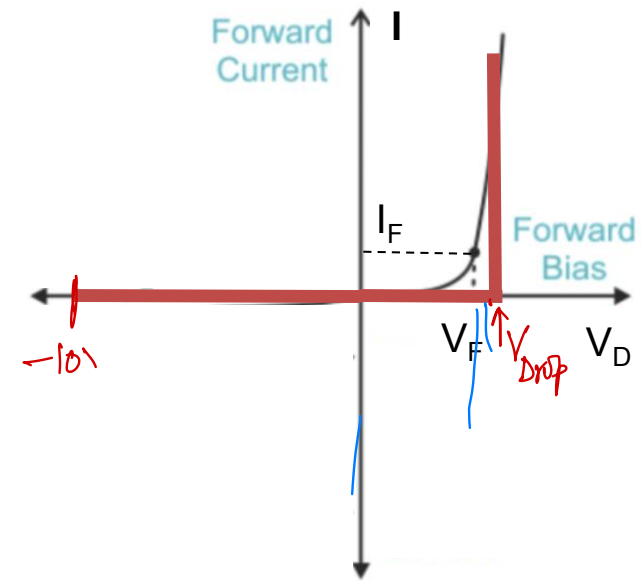


$$V_D = V_{\text{drop}} \sim 0.7 \text{ V}$$

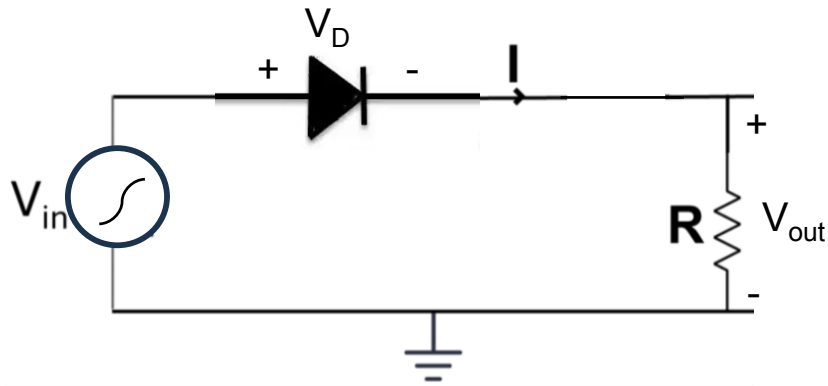
Analysing the diode circuit with AC input. Calculate the output voltage across the resistor.



$$\begin{aligned} V_{\text{out}} &= I R \\ &= \frac{V_{\text{in}} - V_D}{R} \cdot R \\ &= \underline{V_{\text{in}}} - V_{\text{drop}} \end{aligned}$$

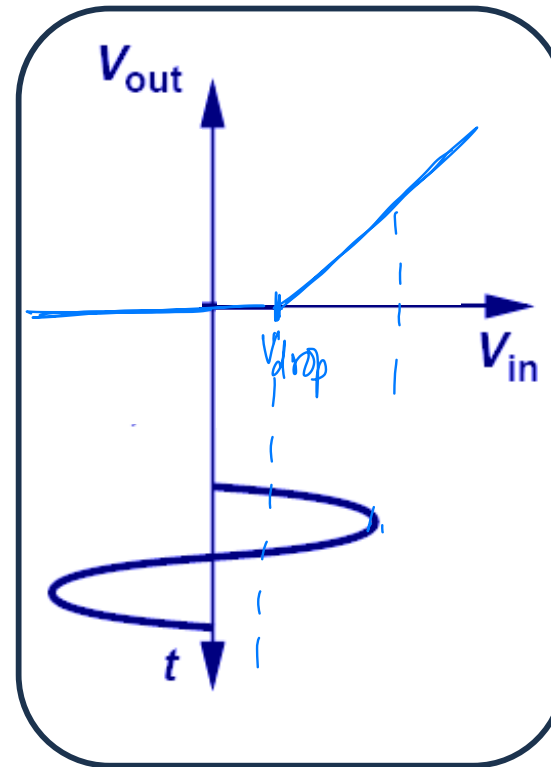
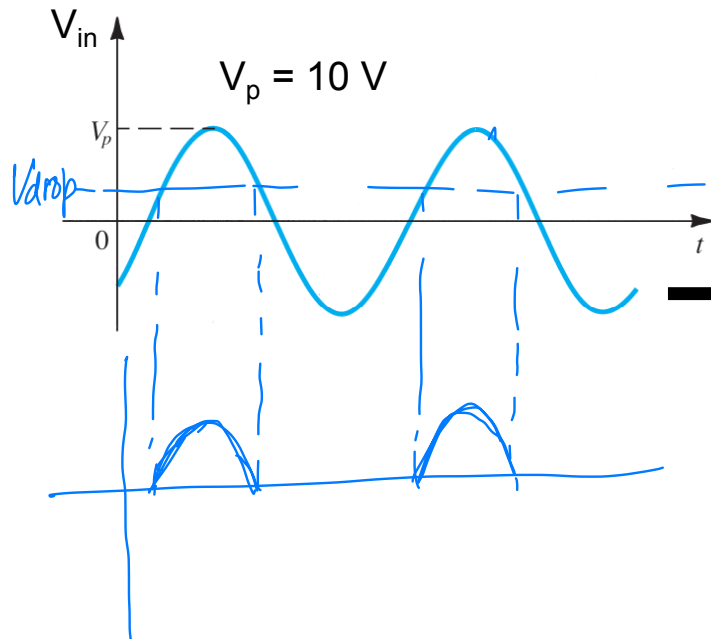


Drawing Circuit's Input-Output Characteristics



$$V_D = V_{drop} \sim 0.7 \text{ V}$$

Analysing the diode circuit with AC input through drawing input-output characteristics (V_{out} vs V_{in}) of the circuit



$$V_{out} = V_{in} - V_{drop} \text{ for } V_{in} > V_{drop}$$

$$V_{out} = 0, \text{ for } V_{in} < V_{drop} \text{ as } I \Rightarrow 0$$